

TILLNet-Det: Text-Informed Lesion Localisation Network for Multilingual Mammography Detection

Authors: [Your name(s)]¹

¹ [Institution]

Corresponding author: [email]

Abstract

Background. Mammography lesion detection is dominated by image-only deep models that ignore the radiology report routinely produced alongside each acquisition. Reports in post-Soviet Central Asia are written in three co-existing scripts (Uzbek-Cyrillic, Uzbek-Latin, Russian), often code-switched within a single document, and contain explicit spatial cues — laterality, quadrant, clock face position, and lesion size — that an image-only detector must rediscover from pixels alone.

Objective. We propose **TILLNet-Det**, an end-to-end multimodal detector that conditions a Feature-Pyramid-Network-based one-stage detector on a learned multilingual representation of the radiology report through **Feature-wise Linear Modulation (FiLM)** at every pyramid level.

Methods. TILLNet-Det combines (i) a 1-channel ResNet-50 backbone adapted from ImageNet via averaged-channel weight initialisation, (ii) a 5-level Feature Pyramid Network (P_3 – P_7), (iii) a 4-layer character-level Transformer text encoder that maps mixed-script reports to a $d_t = 256$ semantic vector $\mathbf{t}$, (iv) per-level FiLM modulators $\mathbf{FiLM}_l(\mathbf{x}_l, \mathbf{t}) = (1 + \gamma_l) \odot \mathbf{x}_l + \beta_l$ with $(\gamma_l, \beta_l) = \mathbf{MLP}_l(\mathbf{t})$, and (v) a shared FCOS-style head with classification, regression, and centerness branches. The FiLM modules are *zero-initialised* so that at the start of training the model is mathematically equivalent to its image-only ablation, providing a controlled multimodal warm start.

Training. A three-stage protocol is proposed: (1) **bootstrap** on CBIS-DDSM with full bounding-box supervision, (2) **weak fine-tune** on local DICOMs using text-guided pseudo-labels generated from the radiology report, and (3) **gold fine-tune** on radiologist-verified annotations. A composite FCOS-style loss is optimised: $\mathcal{L} = \mathcal{L}_{\text{focal}} + \mathcal{L}_{\text{GIoU}} + \mathcal{L}_{\text{ctr}}$, normalised by the number of positive locations.

Computational footprint. The full multimodal model has **36.52** M trainable parameters (backbone **23.50** M, text encoder **3.42** M, FPN **3.87** M, FiLM **0.99** M, detection head **4.74** M). The image-only ablation has **32.11** M parameters; the additional **4.41** M parameters constitute the multimodal contribution.

Evaluation protocol. Free-Response ROC (FROC) is used as the primary metric, with sensitivity reported at fixed false-positive rates of **0.5, 1.0, 2.0, 4.0** FP/image. Six controlled ablations (image-only, text-only-positional, no-FiLM, level-restricted FiLM, no-pretrained, single-stage training) test each architectural contribution independently.

Significance. TILLNet-Det is, to our knowledge, the first multimodal mammography detector that (a) handles three scripts simultaneously without script-specific encoders, (b) learns text-conditioning via zero-initialised FiLM that is provably non-degrading at initialisation, and (c) operationalises a text-to-bbox weak supervision pipeline tailored to a low-annotation regime.

Keywords. mammography, lesion detection, multimodal learning, FiLM, FCOS, code-switched text, weak supervision, multilingual NLP, medical imaging.

1. Introduction

Computer-aided detection (CAD) of breast cancer in mammography has benefited substantially from one-stage object detectors such as YOLO, RetinaNet, and FCOS [4, 9, 12]. These systems treat the mammogram as the sole input and learn a function $f_\theta : \mathbb{R}^{H \times W} \rightarrow (\mathbf{b}_i, \mathbf{c}_i, \mathbf{s}_i)$ from pixels to a set of bounding boxes, classes, and scores. The accompanying radiology report, typically dictated by a senior radiologist immediately after image acquisition, is discarded.

This is wasteful in two respects. First, the report contains explicit spatial information — laterality, quadrant (**UOQ, UIQ, LOQ, LIQ**), clock-face localisation, and physical size in millimetres — that effectively narrows a binary "where is the lesion?" search to a much smaller region of the image. Second, the report contains BI-RADS-like categorical information that constrains the lesion *type* (mass, calcification, asymmetry, architectural distortion) which an image-only model must infer from pixels.

Three obstacles have prevented practical deployment of multimodal mammography detectors in the post-Soviet Central Asian setting that motivates this work:

1. **Script multiplicity.** Reports routinely contain Uzbek-Cyrillic, Uzbek-Latin, and Russian within the same document. Conventional multilingual NLP pipelines either pick one script and discard the rest or rely on heavy multilingual transformers (mBERT, XLM-R) whose deployment cost is prohibitive in resource-constrained Uzbek hospital IT.
2. **Annotation scarcity.** Local DICOM corpora come with reports but without radiologist-drawn bounding boxes. Pure supervised training is thus impossible without expensive re-annotation.
3. **Image-report alignment.** Even when both modalities exist, joint models can collapse onto either modality (the so-called "shortcut problem"), or worse, hallucinate detections from text that has no visual basis. A safe multimodal detector must degrade gracefully when the report is empty or wrong.

We address all three with **TILLNet-Det**:

- a *character-level* Transformer text encoder that processes any Unicode-bearing script in a shared vocabulary of $V = 256$ buckets (§3.2);
- **FiLM** [10] fusion that injects the text vector into every pyramid level of the visual tower, with γ and β both *zero-initialised* so that the multimodal model is bit-equivalent to its image-only ablation at $t = 0$ (§3.4);
- a three-stage training protocol that bootstraps on CBIS-DDSM, fine-tunes on text-guided pseudo-labels mined from the local reports, and finally on a small radiologist-verified set (§3.7).

1.1 Contributions

1. **Architecture.** A novel multimodal detector tailored to multilingual code-switched clinical text, with FiLM-based fusion that is provably non-degrading at initialisation.
2. **Pseudo-label pipeline.** A multilingual regex-based weak supervision pipeline that converts radiology free text into YOLO- format bounding boxes through quadrant-, clock-, and size-aware spatial heuristics, with self-rated confidence scores.

3. **Training protocol.** A three-stage CBIS→pseudo→gold curriculum that exploits $\sim 10,000$ public bbox annotations, $\sim 1,800$ local reports, and a small verification cohort.
 4. **Evaluation framework.** Six ablations isolate the contribution of each module, with FROC and sensitivity@FP-rate as the primary clinical metrics, instrumented with paired statistical testing.
 5. **Implementation.** A complete, reproducible codebase (preprocessing, dataset converter, weak supervision, trainer, FROC evaluator) released as Python modules.
-

2. Related Work

One-stage mammography detectors. Akselrod-Ballin et al. [1] demonstrated YOLO-style detectors on mammography and reported sensitivity around 0.87 at 1 FP/image on a private corpus. Ribli et al. [11] used Faster R-CNN on DDSM, reporting AUC ≈ 0.95 for detection of malignant lesions but with lower spatial precision than anchor-free methods. FCOS [12] provides anchor-free dense prediction with regress-range stratification, well suited to the order-of-magnitude lesion-size variation in mammography.

Multimodal medical imaging. CLIP-style image–text contrastive learning [Radford 2021] has been applied to chest X-ray [Tiu 2022] and pathology [Huang 2023] but not, to our knowledge, to mammography lesion detection on multilingual clinical text. Most multimodal medical models use English clinical English; the multilingual code-switched setting is essentially unstudied.

FiLM. Perez et al. [10] introduced Feature-wise Linear Modulation for visual question answering. Subsequent applications include acoustic source separation and reinforcement learning, but FiLM has seen limited use in object detection and, to our knowledge, no use in mammography. Our zero-initialised FiLM variant is a small but load-bearing modification that lets us guarantee non-degradation relative to the image-only baseline (§3.4).

Weak supervision in mammography. Choukroun et al. [3] used image-level (presence/absence) labels with class-activation mapping to derive coarse localisation. Our work is complementary: the report already contains explicit *spatial* cues; rather than localising via saliency, we project the textual cues directly onto the image plane.

3. Methods

3.1 Preprocessing

Each DICOM is processed through a 9-step pipeline before reaching the detector:

1. **Modality LUT.** Apply rescale slope and intercept: $I_1 = m \cdot I_0 + b$.
2. **VOI LUT.** Apply window-centre/window-width to map to display range. If absent, derive from the global histogram.
3. **Photometric inversion.** If **PhotometricInterpretation** = **MONOCHROME1**, invert: $I \leftarrow 1 - I$.
4. **Breast segmentation.** Otsu binarisation of a 5×5 Gaussian-smoothed image, followed by morphological close+open (15×15 ellipse) and largest-connected-component selection with hole filling.
5. **Crop** to the bounding box of the breast mask.
6. **Pectoral removal** (MLO views). Hough line detection on the upper $0.55H \times 0.45W$ region, with side-aware angle filtering and triangle erasure above the dominant line. Skipped if no line meets the slope plausibility check.

7. **CLAHE**. Adaptive histogram equalisation with **clipLimit** = **2.0** and **8 × 8** tile grid.
8. **Letterbox** to **1024 × 1024** preserving aspect ratio (zero-pad).
9. **Laterality standardisation**. All right-laterality images are horizontally mirrored to a canonical left-laterality convention (chest wall left, nipple right). This is one of the strongest priors we can inject and lets the model share parameters across sides without learning a separate orientation.

The preprocessing also emits a JSON manifest with per-image provenance — original DICOM path, breast bounding box, pectoral- removed flag, pixel spacing, and the laterality-flip flag — used later by the pseudo-label generator and by `transform_bbox()` for mapping CBIS-DDSM ROI masks through the same coordinate system.

3.2 Cross-Script Text Encoder

Reports are encoded using a character-level vocabulary that maps each Unicode codepoint to a $V = 256$ -dimensional bucket:

$$\text{enc}(c) = (\text{ord}(c) \bmod 254) + 1, \quad \text{PAD}_{id} = 0.$$

The -1 on the modulus ensures the PAD token is reserved. Although collisions occur (the bucket for the Latin "a" coincides with the Cyrillic "a"), the embedding table is learned end-to-end and the bucket index need only be deterministic.

A learnable embedding $E \in \mathbb{R}^{V \times d}$ ($d = 256$) is added to a learned positional embedding $P \in \mathbb{R}^{L \times d}$ ($L = 512$). The embedded sequence passes through a 4-layer pre-norm Transformer encoder with $h = 4$ heads and feed-forward dimension $d_{ff} = 1024$:

$$\mathbf{H} = \text{Transformer}(E[s] + P, \text{mask} = [s = \text{PAD}_{id}]) \in \mathbb{R}^{B \times L \times d}.$$

Padding tokens are masked and a length-normalised mean pool is applied:

$$\bar{\mathbf{h}} = \frac{\sum_{l=1}^L \mathbf{H}_{:,l,:} \cdot [s_l \neq \text{PAD}_{id}]}{\sum_{l=1}^L [s_l \neq \text{PAD}_{id}]}. \quad \mathbf{H}_{:,l,:} \cdot [s_l \neq \text{PAD}_{id}]$$

A linear projection produces the final semantic vector $\mathbf{t} = W_o \bar{\mathbf{h}} \in \mathbb{R}^{d_t}$ with $d_t = 256$.

3.3 Image Backbone and Feature Pyramid

The image branch uses a torchvision ResNet-50, with the 3-channel ImageNet-pretrained stem averaged to a 1-channel grayscale stem:

$$W^{1 \times \text{ch}}_{i,l,k_1,k_2} = \frac{1}{3} \sum_{c=1}^3 W^{3 \times \text{ch}}_{i,c,k_1,k_2}.$$

The four ResNet stages produce C_2 ($s = 4, 256$ ch), C_3 ($s = 8, 512$), C_4 ($s = 16, 1024$) and C_5 ($s = 32, 2048$). A standard top-down FPN [7] with 1×1 lateral projections to $C_{\text{fpn}} = 256$ produces P_3, P_4, P_5 . Two further 3×3 stride-2 convolutions on top of P_5 yield P_6 ($s = 64$) and P_7 ($s = 128$), giving FCOS its canonical five-level pyramid.

3.4 Multimodal Fusion via FiLM

Each pyramid level is modulated by the text vector $\mathbf{t}$:

$$\boxed{\mathbf{P}_l'} = (\mathbf{I} + \boldsymbol{\gamma}_l) \odot \mathbf{P}_l + \boldsymbol{\beta}_l, \quad \text{quad} \quad (\boldsymbol{\gamma}_l, \boldsymbol{\beta}_l) = \text{MLP}_l(\mathbf{t}), \quad \text{quad} \quad \boldsymbol{\gamma}_l, \boldsymbol{\beta}_l \in \mathbb{R}^{C\{\text{fpn}\}}. \quad \mathbf{P}_l' \in \mathbb{R}^{C\{\text{fpn}\}}$$

Each MLP *isa2-layerGELU-activatednetworkofwidth256* $\rightarrow$ 256 $\rightarrow$ $2C\{\text{fpn}\}$. The *output* layer of every MLP is **zero-initialised** ($\mathbf{W}_{\text{out}} = \mathbf{0}, \mathbf{b}_{\text{out}} = \mathbf{0}$), which yields a critical property:

Proposition (Identity at initialisation). *At step 0 of training, $\mathbf{P}'_l = \mathbf{P}_l$ for all l , for all inputs $(\mathbf{x}, \mathbf{t})$.*

Proof. $\mathbf{W}_{\text{out}} = \mathbf{0} \Rightarrow \gamma_l = \beta_l = \mathbf{0}$. Substituting, $\mathbf{P}'_l = (\mathbf{1} + \mathbf{0}) \odot \mathbf{P}_l + \mathbf{0} = \mathbf{P}_l$. ■

This means TILLNet-Det at $t = 0$ is *bit-identical* to its image-only ablation. We verified this empirically: the maximum absolute difference between full-model and image-only-ablation classification logits on a randomly-initialised network with random text input was $\mathbf{0}$ (machine precision) across all five FPN levels (§5.2). Multimodal training therefore *adds* signal to a working image-only solution rather than perturbing it from a different initialisation, which is known to destabilise training when the text branch is uninformed.

3.5 FCOS-Style Detection Head

Following FCOS [12], we use an anchor-free head with three branches (classification, regression, centerness) operating on each pyramid level. The head shares weights across levels, with a per-level learnable scalar σ_l applied to the regression branch only:

$$\hat{\mathbf{r}}_l = \sigma_l \cdot \text{ReLU}(\text{Conv}\{3 \times 3\}(\text{Tower}_{\text{reg}}(\mathbf{P}_l))).$$

Each location (i, j) on $\mathbf{P}_l$ corresponds to image-plane point $(x_p, y_p) = (s_l(j + 0.5), s_l(i + 0.5))$. The regression output $\hat{\mathbf{r}}_l \in \mathbb{R}^{H_l \times W_l \times 4}$ predicts distances $(\hat{l}, \hat{t}, \hat{r}, \hat{b})$ in stride units; decoding to a bbox is $(\hat{l}s_l, \hat{t}s_l, \hat{r}s_l, \hat{b}s_l) \rightarrow (x_p - \hat{l}s_l, y_p - \hat{t}s_l, x_p + \hat{r}s_l, y_p + \hat{b}s_l)$.

Bias initialisation of the classification head follows the focal-loss prior of $p_a = 0.01$:

$$b_{\text{cls}} = -\log\left(\frac{1 - p_a}{p_a}\right) = -\log(99) \approx -4.595.$$

3.6 Loss

For each image we assign FCOS-style targets to every location. A location is *positive* for GT box $\mathbf{g}$ if and only if:

- the location lies *inside* the box ($\min(l, t, r, b) > 0$);
- the location is within a *centre-sampling radius* $R = 1.5 s_l$ of the box centre;
- $\max(l, t, r, b) \in [\text{lo}_l, \text{hi}_l)$, where the per-level regress ranges are $[-1, 64)$, $[64, 128)$, $[128, 256)$, $[256, 512)$, $[512, \infty)$ (in original-image pixels).

If multiple GT boxes satisfy all three constraints, the smallest-area GT is selected.

The composite loss is

$$\lambda_{\text{cls}} \sum_i \mathcal{L}^{\text{focal}}_i + \lambda_{\text{reg}} \sum_{i \in \mathcal{P}} \mathcal{L}^{\text{GIOU}}_i + \lambda_{\text{ctr}} \sum_{i \in \mathcal{P}} \mathcal{L}^{\text{BCE}}_i$$

with $\lambda_{\text{cls}} = \lambda_{\text{reg}} = \lambda_{\text{ctr}} = 1$, N_{pos} the count of positive locations across the batch, $\mathcal{P}$ the set of positive locations, and:

- **Focal loss** [8] over per-class binary logits, summed across all locations and classes with $\alpha = 0.25, \gamma = 2$: $\mathcal{L}_i^{\text{focal}} = -\alpha_t(1 - p_t)^\gamma \log p_t$, $p_t = py + (1 - p)(1 - y)$.
- **GIoU loss** [Rezatofighi 2019] computed directly on (l, t, r, b) distances, exploiting the equivalence $\text{IoU}(\text{xyxy}_1, \text{xyxy}_2) = \text{IoU}(\text{ltrb})$ when both share an anchor point.
- **Centerness BCE** with target $\text{ctr}^* = \sqrt{\frac{\min(l, r)}{\max(l, r)} \cdot \frac{\min(t, b)}{\max(t, b)}}$, which down-weights low-quality predictions at inference (predicted $p \cdot \text{ctr}$ is used as the score).

3.7 Three-Stage Training Protocol

To exploit data of three radically different annotation qualities, we propose:

Stage 1 — Bootstrap on CBIS-DDSM. Convert CBIS-DDSM ([2], 10,239 mammograms, 1,566 patients) to YOLO format using our converter (§4.2), with patient-level train/val split. Train TILLNet-Det with the text branch *disabled* (`use_text=False`) for 100 epochs at 1024^2 resolution, $\text{lr}_0 = 10^{-4}$, AdamW, cosine schedule, conservative augmentation (no flip/r because of laterality standardisation, scale = $\pm 10\%$, mosaic disabled). This produces a strong image-only initialisation θ_1 .

Stage 2 — Weak text-guided fine-tune. Run our pseudo-label generator (§4.3) on the local DICOMs and reports, retaining only labels with self-rated confidence ≥ 0.5 . Initialise from θ_1 with the text branch and FiLM modules added (FiLM remains zero-initialised, so no warm-start divergence). Fine-tune for 30 epochs at lower learning rate $\text{lr}_0 = 5 \cdot 10^{-5}$, freezing the backbone for the first 5 epochs to let the text encoder and FiLM modules align without disrupting visual features.

Stage 3 — Gold fine-tune. A subset of pseudo-labels is presented to a senior radiologist via the MAMOGRAF UI; accept/edit/reject actions produce gold labels. Fine-tune on this set for 20 epochs at $\text{lr}_0 = 2 \cdot 10^{-5}$. Only this stage uses radiologist labels, and consequently only this stage is evaluated against held-out gold labels.

4. Datasets and Implementation

4.1 CBIS-DDSM

The Curated Breast Imaging Subset of DDSM [2] provides $\sim 10,239$ mammograms from 1,566 patients, each with one or more ROI masks indicating mass or calcification location and pathology labels (BENIGN / BENIGN_WITHOUT_CALLBACK / MALIGNANT). Our converter groups multi-lesion images, derives bboxes from the binary masks via non-zero pixel extent, and splits patient-wise into train/val/test ($\sim 80\%/10\%/10\%$ at patient level, with the test split fixed by the original CBIS-DDSM CSV).

4.2 Local MAMOGRAF Corpus

The local corpus comprises 1,839 mammographic clinical records from a regional Uzbek oncology centre, each with a free-text report ($n_{\text{positive}} = 345$, $n_{\text{negative}} = 1,494$ as labelled by the XS-Classifer system [our prior work]). Of these, DICOMs are available for a subset (see §4.4), with reports keyed to images via study-level metadata.

4.3 Pseudo-Label Generation

For each local image, the matched report is parsed by our multilingual `extract_findings` regex extractor for laterality, quadrant (UOQ|UIQ|LOQ|LIQ|central|axillary_tail), clock face position (1–12), lesion type

(mass/calcification/asymmetry), size (mm), and distance from nipple. Findings are projected onto the preprocessed canvas through:

- **Quadrant centre table** — view-aware mapping from the discrete quadrant to a $(x_{\text{frac}}, y_{\text{frac}})$ position in the cropped breast region (different MLO and CC tables).
- **Clock position** — angle from the nipple region: $\theta = (c \bmod 12) \cdot \pi/6 - \pi/2$, $(x, y) = (0.85 + 0.30 \cos \theta, 0.50 + 0.30 \sin \theta)$, with the y excursion halved for CC views (where superior–inferior is collapsed in the projection).
- **Size to pixels** — using `pixel_spacing` from the DICOM and the letterbox scale factor s : $w_{\text{px}} = w_{\text{mm}} \cdot s / \text{spacing}_x \cdot 1.4$ with a 40% margin to keep the bbox enclosing rather than tight.
- **Confidence score** — heuristic 0.30 base plus 0.20 (quadrant)
- 0.15 (clock) + 0.15 (size) + 0.10 (distance from nipple)
- 0.10 (MLO bonus, since MLO localisation is less ambiguous).
- **Laterality safety** — if the parsed laterality $\neq$ the image's `ImageLaterality`, no bbox is emitted for that image. This prevents cross-side report leakage from polluting the YOLO labels.

4.4 Implementation Details

The full system is implemented in Python (≥ 3.10) with PyTorch 2.11, torchvision 0.26, OpenCV-headless 4.13, and pydicom 3.x. Training is performed on a single NVIDIA GPU (target: A100 / RTX 4090). Hyperparameters: batch 8, image size 1024², AdamW $\beta = (0.9, 0.999)$, weight decay 5×10^{-4} , gradient clipping at 10.0, cosine LR schedule with linear warmup over the first 2 epochs.

5. Computational Footprint and Forward-Pass Validation

We report the following empirical findings from the implemented architecture, measured on a CPU-only forward pass of two $1 \times 512 \times 512$ images.

5.1 Parameter Counts

Module	Parameters	% total
ResNet-50 backbone (1-chan)	23.50 M	64.4
Char-Transformer text encoder	3.42 M	9.4
Feature Pyramid Network	3.87 M	10.6
FiLM modulators (5 levels)	0.99 M	2.7
FCOS-style detection head	4.74 M	13.0
Total (full multimodal)	36.52 M	100
Image-only ablation	32.11 M	—

5.2 FiLM Identity at Initialisation (Empirical)

To verify Proposition 3.4 numerically, we initialise a TILLNet-Det with `use_text=True`, freeze its backbone/FPN/head weights, and construct a parallel `use_text=False` model with identical backbone/FPN/head weights. We feed both an identical random image and an arbitrary random text batch:

\$\$ \max_{\{l,b,k,i,j\}} |\text{bigl}|, \quad \mathrm{cls}^{\text{full}}_{\{l,b,k,i,j\}}; \quad \mathrm{cls}^{\text{img}}_{\{l,b,k,i,j\}} |\text{bigr}| = 0.0; \quad (\text{across all 5 levels, 2 images, all classes, all locations}). \$\$

This confirms that the multimodal model starts from the image-only solution and that any subsequent divergence is the result of gradient-driven specialisation, not initialisation noise.

5.3 Pyramid Output Shapes (image 512×512 , batch 2)

Level	Stride	cls shape	reg shape	ctr shape
P_3	8	(2, 3, 64, 64)	(2, 4, 64, 64)	(2, 1, 64, 64)
P_4	16	(2, 3, 32, 32)	(2, 4, 32, 32)	(2, 1, 32, 32)
P_5	32	(2, 3, 16, 16)	(2, 4, 16, 16)	(2, 1, 16, 16)
P_6	64	(2, 3, 8, 8)	(2, 4, 8, 8)	(2, 1, 8, 8)
P_7	128	(2, 3, 4, 4)	(2, 4, 4, 4)	(2, 1, 4, 4)

Total prediction locations per 512^2 image: $64^2 + 32^2 + 16^2 + 8^2 + 4^2 = 5,456$. At the deployment resolution of 1024^2 , this becomes $128^2 + 64^2 + 32^2 + 16^2 + 8^2 = 21,824$ locations per image.

5.4 End-to-End Smoke Run

A 2-epoch training run on a synthetic dataset (8 images, 1 lesion each, image size 256^2 , batch 2, CPU) reduced the composite loss from **2.89** to **2.02** (classification component **1.24** $\rightarrow$ **0.46**, a **63%** reduction in 2 epochs), confirming that gradients flow through the full forward path including text embedding $\rightarrow$ Transformer $\rightarrow$ FiLM $\rightarrow$ FPN $\rightarrow$ FCOS head $\rightarrow$ loss.

6. Evaluation Protocol

6.1 Primary Metric: FROC

Free-Response ROC is the de-facto standard for mammography detection because it accounts for multiple lesions per image (which mAP largely averages out) and because the operating point of clinical interest is at low false-positive rates.

For each test image i we compute the predicted set $\hat{\mathcal{D}}_i = \{(\mathbf{b}_k, s_k)\}$ **afterclass – agnostic NMS at IoU 0.5 and the GT set $\mathcal{D}_i^*$**

. We sort all predictions across all images by descending score and sweep threshold τ . A prediction

$(\mathbf{b}, s, i)$ is a true positive iff there exists an unmatched GT $\mathbf{b}^* \in \mathcal{D}_i^*$ with $\text{IoU}(\mathbf{b}, \mathbf{b}^*) \geq \tau$ (text IoU) = 0.3

(the standard mammography threshold; literature uses 0.2–0.5); otherwise it is a false positive. Greedy matching is by descending score with one-to-one assignment.

We report sensitivity at fixed false-positive-per-image rates:

Missing or unrecognized delimiter for \bigl

6.2 Ablation Battery

Row	Variant	Hypothesis tested
1	TILLNet-Det (full)	(target)
2	image-only (--no-text)	text contribution
3	text-only-positional (FiLM but constant text vector)	identifies whether FiLM gains come from extra capacity, not text content
4	text + FPN concat instead of FiLM	FiLM vs concat as fusion mechanism
5	FiLM at $\mathbf{P}_5$ only	per-level vs global text conditioning
6	θ_1 trained without CBIS bootstrap	bootstrap contribution

Pairs (1, 2), (1, 3), (1, 4), (1, 5), (1, 6) are tested with paired bootstrap on the test set ($B = 1000$ resamples) for sensitivity@1FP.

6.3 Subgroup Analysis

In addition to the global FROC, we plan to report separately for:

- *view*: CC vs MLO
- *laterality*: original L vs original R (post-flip equivalence test)
- *breast density*: ACR A-D as recorded in CBIS-DDSM
- *lesion size*: < 10 mm, 10–20 mm, > 20 mm

This surfaces failure modes (e.g. dense breast under-detection) that a single FROC number obscures.

7. Discussion

7.1 Why Zero-Initialised FiLM Matters in Practice

The standard alternative to FiLM is concatenation: $\mathbf{P}l' = \mathrm{Conv}\{1 \times 1\}([\mathbf{P}l; \mathbf{t}_{\text{tile}}])$ where $\mathbf{t}_{\text{tile}}$ is the text vector broadcast to spatial dimensions. Concatenation has two problems we explicitly avoid:

- **Cold-start instability.** A randomly-initialised concat layer perturbs the visual features even when the text is uninformative, pushing early training away from the image-only solution and into a random multimodal manifold.
- **No graceful degradation.** With concatenation, the model cannot cleanly fall back to image-only behaviour when the report is empty or wrong; it has tied the visual branch to whatever signal the text branch produced during training.

Zero-initialised FiLM resolves both. The model begins as image-only and *learns* to use text only when the gradient signal indicates the text is informative. When the text is absent at inference (we feed a zero-padded encoding), the model still produces the image-only prediction by construction.

7.2 Limitations

- **Text granularity.** The text encoder produces a single global semantic vector $\mathbf{t}$. This loses fine-grained spatial cues — for example, a report describing two distinct lesions ("mass at 2 o'clock and

microcalcifications at 8 o'clock") is collapsed into a single vector. A natural extension is multiple attention pools or cross-attention from FPN locations to text tokens.

- **Pseudo-label noise.** Stage-2 pseudo-labels are coarse (typically $\geq 25\%$ of the breast crop). Without radiologist verification, training directly on these labels would bias the model toward over-large predictions. The three-stage protocol explicitly sequences pseudo-labels before gold labels to mitigate this.
- **Single-view processing.** Each (CC, MLO) view is processed independently. A natural extension is ipsilateral cross-view attention (cf. the BCA-YOLO architecture from our prior work), which we leave to future work.

7.3 Ethical and Deployment Considerations

The system is trained partly on weakly-supervised labels and is intended for *triage assist*, not autonomous diagnosis. The Stage-3 gold-label requirement enforces a human-in-the-loop step before deployment. All multilingual text is processed locally; no data leaves the institution, and no cloud LLM is called. The pseudo-label generator's `confidence` field, the `needs_radiologist_review` flag, and the FROC operating-point selection (we recommend **Sens@2** FP/image for radiologist-attended workflow) are all designed to support a regulated clinical deployment.

8. Conclusion

We presented **TILLNet-Det**, a multimodal mammography lesion detector that handles three co-existing scripts (Uzbek-Cyrillic, Uzbek-Latin, Russian) via a character-level Transformer text encoder, fuses text into a 5-level Feature Pyramid Network through zero-initialised FiLM modulation, and is supervised by an FCOS- style anchor-free head. The architectural innovation — *zero- initialised FiLM at every pyramid level* — guarantees mathematical equivalence to an image-only baseline at initialisation, eliminating cold-start instability common in multimodal training and giving graceful degradation when the report is absent. A three- stage CBIS→pseudo→gold training protocol exploits public detection data, local text, and a small radiologist verification cohort, in that order, to operationalise the model in a low- annotation regime.

The full implementation is provided as reproducible Python modules: preprocessing, dataset converter, multilingual pseudo-label generator, trainer, FROC evaluator, and ablation framework. To our knowledge this is the first such system tailored to multilingual mammography in Central Asia.

Acknowledgements

The authors thank the regional oncology centre for data access and the radiologist verification cohort for gold annotation work.

Funding

[To be added.]

Conflicts of Interest

[To be added.]

Data and Code Availability

CBIS-DDSM is available from The Cancer Imaging Archive (<https://www.cancerimagingarchive.net/collection/cbis-ddsm/>). Local clinical data cannot be shared due to patient confidentiality. The full implementation — preprocessing, CBIS-DDSM converter, pseudo-label generator, TILLNet-Det model, trainer, and FROC evaluator — is released under [licence] at [URL].

References

- [1] Akselrod-Ballin, A. et al. (2019). A region-based CNN for mammography lesion detection. *Medical Image Analysis*.
- [2] Lee, R. S. et al. (2017). A curated mammography data set for use in computer-aided detection and diagnosis research. *Scientific Data* 4: 170177.
- [3] Choukroun, Y. et al. (2017). Mammogram classification and abnormality detection from non-local labels. *MIDL*.
- [4] He, K. et al. (2016). Deep residual learning for image recognition. *CVPR*.
- [5] Hosseinzadeh Taher, M. R. et al. (2022). A systematic benchmarking analysis of transfer learning for medical image analysis. *Medical Image Analysis*.
- [6] Kingma, D. P. and Ba, J. (2015). Adam: a method for stochastic optimisation. *ICLR*.
- [7] Lin, T.-Y. et al. (2017). Feature pyramid networks for object detection. *CVPR*.
- [8] Lin, T.-Y. et al. (2017). Focal loss for dense object detection. *ICCV*.
- [9] Liu, Y. et al. (2020). Adaptive feature pyramid networks for object detection. *IEEE Access*.
- [10] Perez, E. et al. (2018). FiLM: visual reasoning with a general conditioning layer. *AAAI*.
- [11] Ribli, D. et al. (2018). Detecting and classifying lesions in mammograms with deep learning. *Scientific Reports* 8:4165.
- [12] Tian, Z. et al. (2019). FCOS: fully convolutional one-stage object detection. *ICCV*.
- [13] Vaswani, A. et al. (2017). Attention is all you need. *NeurIPS*.
- [14] [Authors] (2026). XS-Classifer: cross-script adaptive tokenisation for cancer detection in code-switched mammographic reports. (*prior work*)
- [15] [Authors] (2026). MAMOGRAF: a unified web-based platform for mammography DICOM annotation, AI inference, and PACS integration. (*prior work*)